6.8 Piecewise Functions

Name _____ Period ____ Date _____

Algebra II 6.8 (30-minute plan) · Student sheet

I can…

- evaluate piecewise functions.
- graph and write piecewise functions.
- write step functions and absolute value functions as piecewise functions.

Vocabulary · fill in the missing word

Piecewise function — a function defined by two or more _____, each for a different part of the domain

Step function — a _____ function whose pieces are constant, so the graph looks like stairs

Open / closed circle — the _____ is not included / is included

Part 1 · Notes and examples (fill in each note, then work the example)

Piecewise function: A **piecewise function** uses different _____ on different parts of the _____.

Evaluating: To evaluate, first find which piece's _____ the input fits.

Example 1

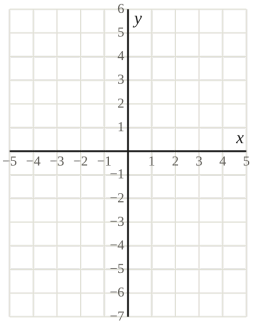
Let $f(x) = 2x + 1$ if $x \le 1$, and $f(x) = x^2 - 3$ if $x > 1$.

Find $f(-2)$, $f(1)$, and $f(3)$.

Graphing: ≤ or ≥ → _____ circle. < or > → _____ circle.

Example 2

Graph $f(x) = x + 3$ if $x < 1$, and $f(x) = -2x + 1$ if $x \ge 1$.



Step function: A **step function** has pieces that are all _____.

Example 3

A parking garage charges $5 for the first hour or part of an hour and $3 for each additional hour or part of an hour, up to 3 hours.

Write a step function for the cost. How much is 2.5 hours?

Part 2 · Partner practice

Partner practice

Let $f(x) = -x$ if $x < -1$; 4 if $-1 \le x \le 2$; x^2 if $x > 2$.

Find $f(-3)$, $f(2)$, and $f(5)$.